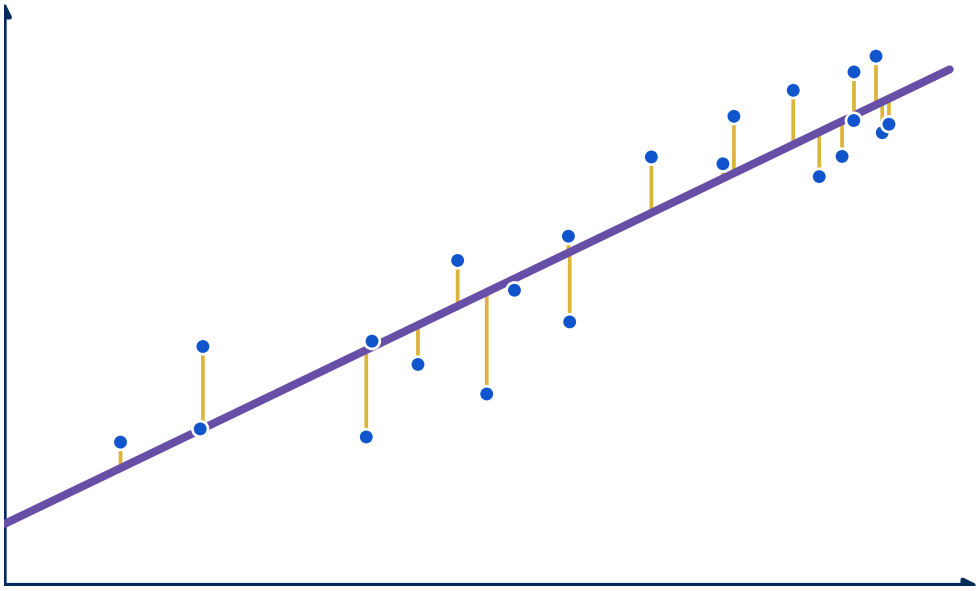


UNIT 02 · CHAPTER 2

Linear Regression

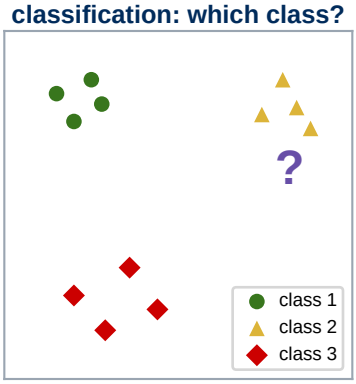
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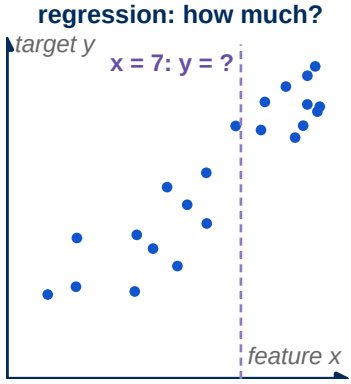
William Theisen · University of Notre Dame

Classification and regression

Our **previous models**, **k-Nearest Neighbors** and **Decision Trees**, were built for **classification**.

A **classification task** is one in which we try to assign a _____ to a sample.





In **regression tasks**, we try and assign a _____ to each sample.

Regression is another form of _____.

Quick check (old exam question): is a regression task supervised or unsupervised learning?

Linear regression

A **linear regression** is a technique to fit a line to our data, allowing us to make predictions.

Regression vocabulary

Unfortunately, because linear regression is so old, it **has a lot of vocab specific to it**:

[2.1] a line with one feature

$$\hat{y} = w_1 x_1 + w_0$$

- $\hat{y}$: _____
- w_1, w_0: _____
- x_1: _____

Matrix notation

We can represent basically everything as **matrices** and **matrix operations**.

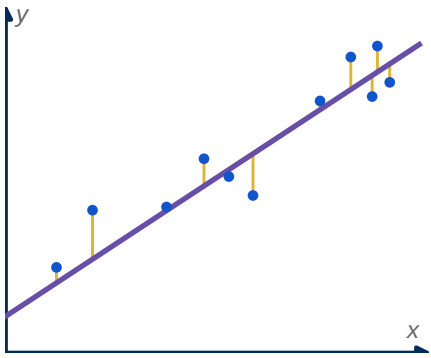
Multiple features

It's rare that we have datasets with only a single feature. The **iris dataset**, a toy dataset, has 4!

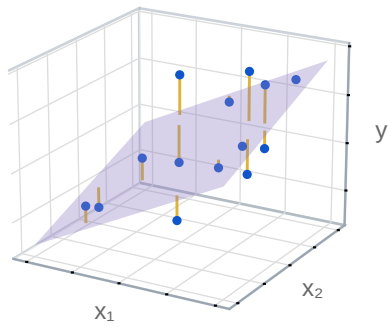
When we perform a **linear regression on** a dataset with **more than one feature**, we call this a _____.

Hyperplanes

1 feature: a line



2 features: a plane

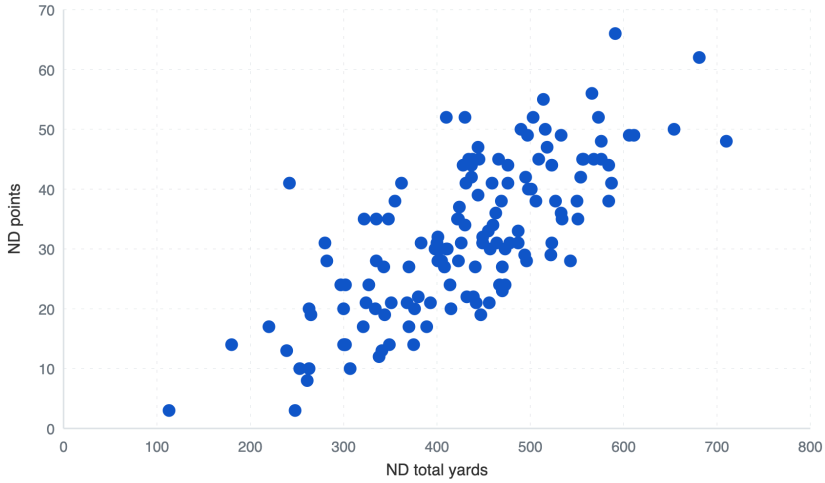


With one feature the model is a line, with two it's a _____ and with more it's a _____.

Quick check (old exam question): what would the equation look like for a multiple linear regression with three features?

Fitting our line

How do we actually find the line that "fits" the best?



Draw the line you think fits best. Which games is it furthest from?

Optimization problems

We need some way to **measure how well our model fits** our data. We can call this a _____.

For **linear regression** we often use **SSE** or ____.

sum of squared errors

$$[2.3] \quad \text{SSE}(\mathbf{w}) = \sum_i (y_i - \hat{y}_i)^2$$

mean squared error

$$[2.4] \quad \text{MSE}(\mathbf{w}) = \frac{1}{n} \sum_i (y_i - \hat{y}_i)^2$$

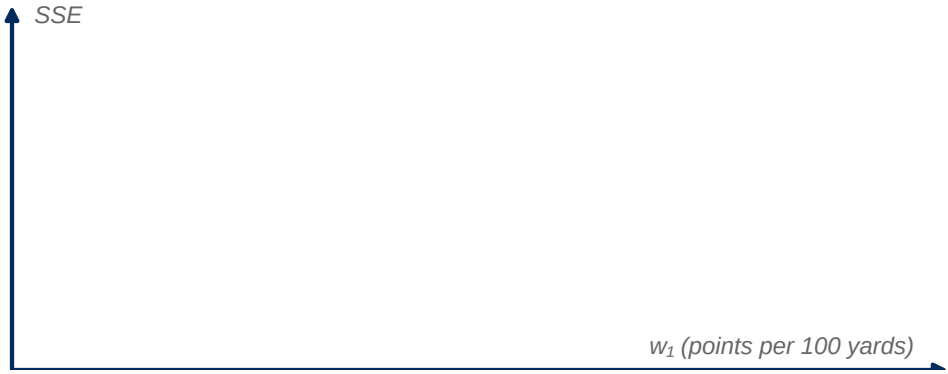
Finding our weights

So given a **loss function**, our **goal** is to _____ it.

When would a **loss function** be **minimized**? When the _____ is ____.

Visualizing loss

We can plot the loss function with respect to **w**:



Ordinary least squares

We can **solve for the gradient equaling 0**:

This is called **Ordinary Least Squares (OLS)**.

Worked example: OLS by hand

game	yards	x_1	y (points)
2019 Michigan	180	2	14
2012 Michigan State	300	3	20
2013 Purdue	400	4	31
2019 Boston College	501	5	40
2017 Temple	606	6	49

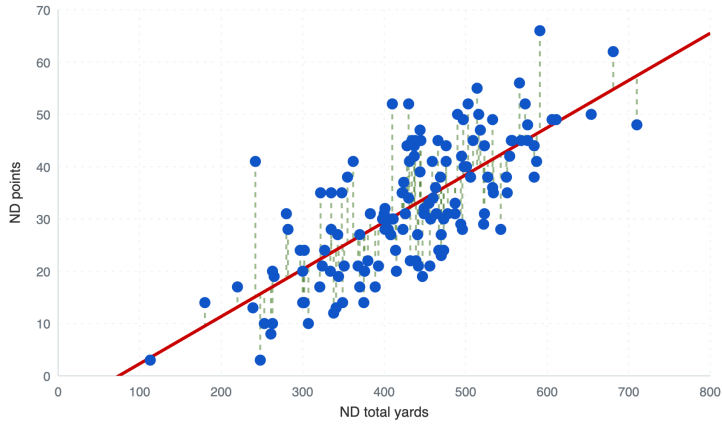
Give X a **column of 1s** for the intercept, then follow the **OLS** recipe.

Reading the weights

So $\hat{y} = 9.0\,x_1 - 5.2$: about _____.

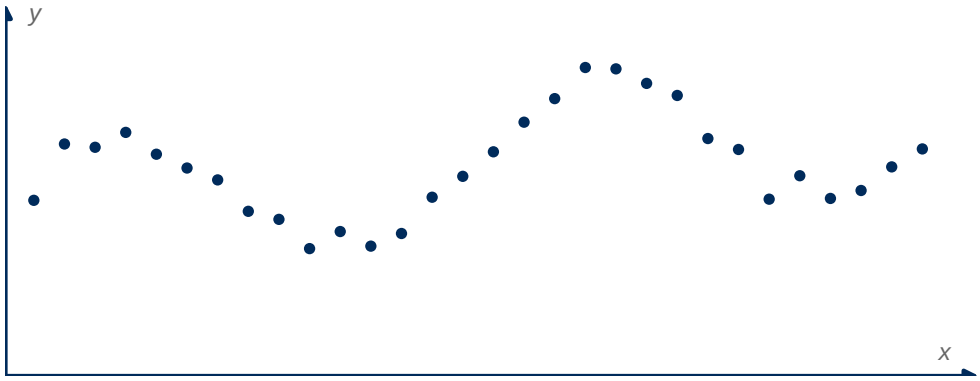
Quick check: in 2018 Notre Dame gained 550 yards against Stanford. What does this line predict, and what is the residual if Notre Dame scored 38?

Linear regression visualization



Representing curves

How could you use a **linear regression** to model a sine wave?



The model only has to be linear in its _____, not in its inputs.

A possible issue

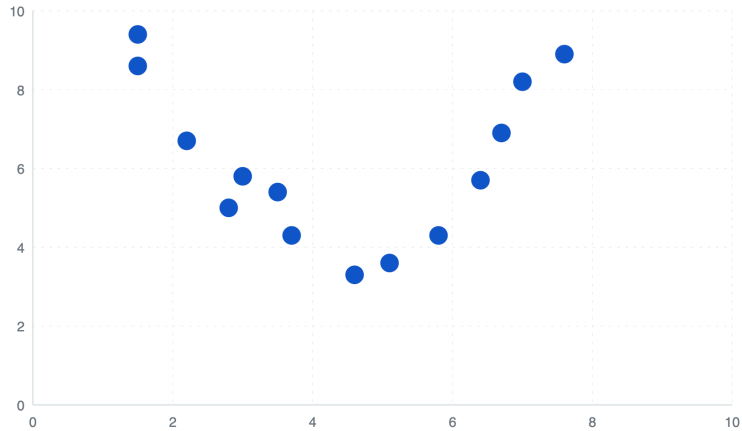
The **weight calculation relies on computing the inverse**, which can be **extremely computationally expensive** and **sometimes just literally impossible**.

Maximum likelihood estimation

Maximum likelihood estimation (MLE) is a method of estimating the parameters of an _____, **given some observed data**.

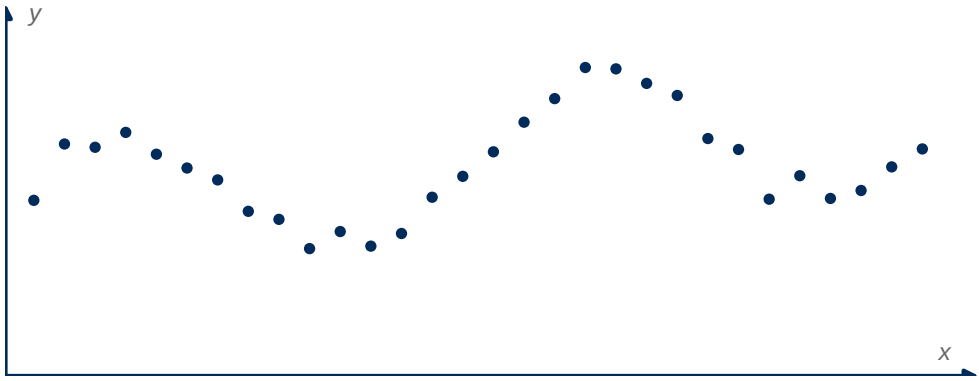
Overfitting

Unfortunately, **Linear Regressions** (a form of MLE), are **prone to overfitting**.

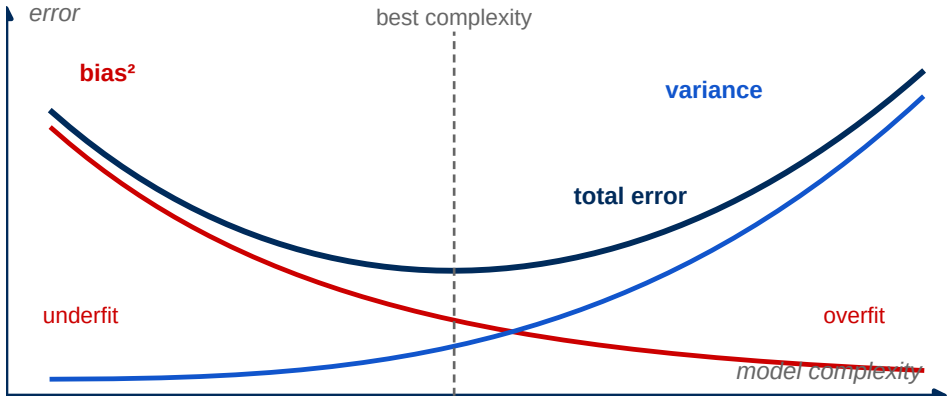


Underfitting

Underfitting is when your model is "too weak" to capture all of the underlying patterns in the data.



Bias and variance



The **bias–variance tradeoff** describes the **relationship between** a **model's complexity**, the **accuracy** of its predictions, and how well it can make **predictions on unseen data** that were not used to train the model.

Practice: an old exam question

x_1	x_2	y
1	1	9
2	2	25
3	3	50
4	4	84
5	5	127

Given the dataset above, answer the following questions **(Old exam question)**:

A. Would a **purely linear model** exhibit **high bias** here or **high variance**? Would it be **overfit** or **underfit**? Explain your reasoning.

B. Design a regression equation that would be able to approximate the dataset.

C. What role does the **SSE** play in **linear regression**? Why does it matter that our resulting parameter space is convex?

D. Look at the x_1 and x_2 columns. What goes wrong if we fit $\hat{y} = w_0 + w_1 x_1 + w_2 x_2$ with **OLS**?

Matching: linear regression

Write the letter of the matching definition next to each term.

___ **Ordinary Least Squares**	**A.** The model memorizes noise in the training data instead of the general pattern
___ **Maximum Likelihood Estimation**	**B.** The relationship between model complexity, training accuracy, and accuracy on unseen data
___ **Underfitting**	**C.** Solving for the weights by setting the gradient of the loss equal to 0
___ **Overfitting**	**D.** Estimating the parameters that make the observed data most likely
___ **Bias–variance tradeoff**	**E.** The model is too weak to capture the underlying patterns in the data

Practice problems

1. Units. Our line is $\hat{y} = 9.0\,x_1 - 6.7$ with x_1 in hundreds of yards. A classmate refits it with x_1 in plain yards. What happens to w_1, to w_0, and to the predictions?

2. Reading a plane. With two features our model is $\hat{y} = 8.8\,x_1 - 2.7\,x_2 - 2.3$, where x_2 is Notre Dame's turnovers. In 2014 against North Carolina, Notre Dame gained 516 yards, turned the ball over 3 times and scored 50. What does the model predict, and what is the residual? Then try the 2016 NC State game: 113 yards, 3 turnovers. What's strange about that prediction?

3. Home field. Fitting Notre Dame's margin of victory (its points minus the opponent's) with one feature, $x = 1$ for a home game and 0 for an away or neutral-site game, gives $\hat{y} = 8.7\,x + 6.4$ with $R^2 = 0.06$. What do w_0 and w_1 mean here? Does the tiny R^2 mean home field doesn't matter?

Practice problems, continued

4. Too good to be true. A student predicts Notre Dame's margin of victory using two features, Notre Dame's points and the opponent's points. OLS gives weights of exactly +1 and −1 and $R^2 = 1$. What went wrong?

5. Training error. On the held-out page, the 119-feature model has a training MSE of 0 and a test MSE of 1,443. Without looking at the test set, how could you have guessed it would do badly?